# How Many Labels Before Which Labels? Risk-Budgeted Feedback Acquisition under Selective Labels

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**Abstract.** Lending decisions determine which repayment outcomes become available for future learning. Under a predicted-loss budget, selecting applicants for information value may improve feedback composition while reducing observable outcomes. We study this trade-off with lexicographic risk-budgeted feedback acquisition (LRBE), which computes the maximum feasible exploratory count before optimizing a secondary score. Random-order replay covers four credit datasets and five budget ratios. A five-policy experiment replaces FAVE's linearized score with bootstrap-committee information gain (IG). At $\rho = 0.1$, LRBE-IG acquires 31.3 more labels than QualityFirst-IG on HMEQ, 49.5 on LendingClub, and 40.1 on UCI, but the corresponding Recall intervals include zero; its UCI Brier score is slightly worse. Relative to minimum-risk selection, LRBE-IG uses 28,765.5 more predicted-risk units on HMEQ and 27,025.7 more on LendingClub without a Recall or Brier interval excluding zero. FAVE and randomized controls show a similar distinction between preserving count and choosing composition. Minimum-risk maximum-count selection therefore provides a practical reference: an acquisition score should be evaluated by the feedback it displaces, the predicted risk it consumes, and its downstream predictive return.

**Keywords:** Selective labels · Feedback acquisition · Risk budgets · Credit prediction

## 1 Introduction

Credit decisions determine which data become available for subsequent learning. Repayment outcomes are observed for approved applicants, while the outcomes of rejected applications remain unknown. This selective-label problem can leave evaluation and retraining dependent on the policy that collected the data [14,13]. Acquiring feedback from a previously rejected region therefore entails a lending decision, not simply an annotation request: observing an outcome requires accepting the associated credit exposure.

Existing research addresses this dependence through reject inference [4], decision policies for selective observations [20], optimistic exploration [17], and adaptive bounded debiasing [22]. Recent credit experiments also use controlled

approvals to investigate feedback loops [18]. These approaches motivate a practical question about resource allocation: under a predicted-loss budget, does sacrificing feasible feedback quantity for a higher information score deliver sufficient downstream benefit? Non-uniform acquisition costs are already studied in active learning [8,19]; our focus is the comparison of quantity-preserving and composition-oriented policies when the cost is an applicant-level credit-loss proxy.

We examine this question using lexicographic risk-budgeted feedback acquisition (LRBE). At each round, the first stage computes the maximum feasible count $K_t^*$ under predicted costs $\widehat{c}_{it} = \widehat{p}_{it}\mathrm{EAD}_i\mathrm{LGD}_i$. The second stage optimizes a sample-composition score subject to that count equality. A secondary score can therefore change which applicants are selected without reducing the current maximum count. Removing the equality produces a composition-first policy and exposes the number of feasible observations exchanged for a higher score. Minimum-risk, joint-objective, and randomized policies provide complementary references.

The evaluation combines three views of the trade-off. Dynamic replay on four credit datasets measures complete policy outcomes over five budget ratios. A five-policy experiment at $\rho = 0.1$ compares FAVE with bootstrap-committee information gain (IG), an acquisition score that does not use FAVE's Fisher geometry or influence approximation. Same-state audits then give every selector the same visible set, candidates, costs, and numerical budget, while a controlled one-step experiment varies exposure heterogeneity and the evaluation weight of an underrepresented region. Dynamic policies share their initial data and incoming batches but subsequently retrain on their own feedback, so later candidate sets and budgets may differ.

The experiments reveal an asymmetric trade-off. Composition-first FAVE and IG often give up many feasible observations, yet neither produces a consistent predictive advantage. Preserving count does not guarantee better prediction either: LRBE-IG acquires 31.3, 49.5, and 40.1 more labels than QualityFirst-IG on HMEQ, LendingClub, and UCI, respectively, without a corresponding Recall interval excluding zero, and its UCI Brier score is slightly worse. At nearly the same cumulative count as MinRisk, both FAVE and IG consume more predicted risk without a consistent downstream return. The controlled experiment reproduces the immediate count and risk trade-offs, while its prespecified Brier interactions remain uncertain.

This work contributes an empirical framework for evaluating risk-budgeted acquisition. First, the LRBE comparison separates the current maximum feasible count from the score used to choose among applicants, making the opportunity cost of composition-first selection measurable. Second, matched-score comparisons with FAVE and IG, together with randomized controls, show how that choice changes feedback quantity, predicted-risk use, Recall, and Brier score. Third, same-state and controlled-condition analyses locate the trade-off at the selection step and test whether exposure heterogeneity or target-region emphasis explains its predictive return. The resulting criterion is concrete: compare an

acquisition score with minimum-risk maximum-count selection and report what it displaces, what it costs, and what it improves.

## 2 Related Work

*Learning from selectively observed outcomes.* Work on selective labels studies the evaluation and learning problems created by decision-dependent observation [14,13]. Expert consistency and stochastic fair decisions provide ways to learn or collect data beyond existing acceptance patterns [6,12]. Credit reject inference examines augmentation and sample-selection correction [4,15], while finite-domain policy optimization and one-sided-feedback learning formalize sequential decisions with missing outcomes [20,10]. Our replay retains observed historical outcomes and reveals them according to simulated approvals. It evaluates an acquisition policy, not an imputation method or a field trial on actual rejected borrowers.

*Exploration objectives and costs.* PLOT uses optimistic pseudo-labels to counter self-confirming rejection and gives regret guarantees under stated modeling assumptions [17]. Adaptive bounded exploration trades debiasing against the risk of accepting otherwise rejected individuals [22]. Partial-feedback exploration also supports subgroup coverage and fairness objectives [11], while cost-efficient fairness auditing accounts for assessment, processing, and potential-default costs [5]. Controlled approvals in credit scoring examine the consequences of retraining under selection bias [18]. These are direct precedents for costly feedback acquisition. We instead compare how a current predicted-loss allowance is allocated between feedback quantity and composition. The randomized controls in our experiments are in-house strategy controls, not implementations of these published algorithms.

*Resource constraints and predictive evaluation.* Cost-sensitive active learning already accounts for heterogeneous query costs [8,19]. Contextual bandits with knapsack constraints and conservative exploration place learning within resource or baseline-performance restrictions [1,21]; safety-constrained approaches further consider uncertainty in feasibility [2,16]. LRBE uses a point predicted-loss constraint rather than these sequential guarantees. Its secondary scores are evaluated by the resulting data and predictions, not by score values alone. We therefore compare Recall and Brier score while reporting predicted-budget compliance separately from realized defaulted exposure.

## 3 Risk-Budgeted Feedback Acquisition

### 3.1 Sequential observation and policy-specific budgets

At round $t$, policy $m$ receives an incoming batch $\mathcal{U}_t$ and fits a default predictor on its visible labeled set $\mathcal{L}_t^{(m)}$. An outcome $y_i = 1$ denotes default. Ordinary

approvals have predicted default probability below the fixed threshold $\tau = 0.5$, which serves as a common operating point for every policy in the replay. The remaining applicants form the exploration candidates

$$\mathcal{C}_t^{(m)} = \{i \in \mathcal{U}_t : \widehat{p}_{it}^{(m)} \geq \tau\}. \tag{1}$$

The policy selects exploratory approvals $\mathcal{S}_t^{(m)} \subseteq \mathcal{C}_t^{(m)}$. Only outcomes of ordinary and exploratory approvals are added to the next visible set. Rejected applicants are not carried over to a backlog. Outcomes are revealed within the simulation round; actual repayment delays are not modeled.

Write $a_i$ for exposure and set $\mathrm{LGD}_i = 1$. The planning cost, exploration budget, and selected risk are

$$\widehat{c}_{it}^{(m)} = \widehat{p}_{it}^{(m)} a_i, \qquad B_t^{(m)} = \rho \sum_{i \in \mathcal{C}_t^{(m)}} \widehat{c}_{it}^{(m)}, \qquad \widehat{R}_t^{(m)} = \sum_{i \in \mathcal{S}_t^{(m)}} \widehat{c}_{it}^{(m)}. \tag{2}$$

The constraint $\widehat{R}_t^{(m)} \leq B_t^{(m)}$ covers exploratory approvals only, not the entire lending portfolio. Defaulted exposure is evaluated separately as $R_t^{\mathrm{real},(m)} = \sum_{i \in \mathcal{S}_t^{(m)}} y_i a_i$. With unit LGD this is an experimental loss proxy, not recovery-adjusted net loss or full IFRS 9 accounting.

Policies share the initial split and exogenous batches, but their feedback choices affect subsequent training sets, predictions, candidates and budgets. Pairing on dataset, seed and $\rho$ therefore compares complete policies under common incoming data. It does not fix their subsequent numerical budgets or isolate a causal effect of composition. Superscripts $m$ are suppressed below for within-policy optimization.

### 3.2 Lexicographic quantity and composition

LRBE first determines the maximum feasible exploratory count:

$$K_t^* = \max_{z \in \{0,1\}^{|\mathcal{C}_t|}} \sum_i z_i \quad \text{subject to} \quad \sum_i z_i \widehat{c}_{it} \leq B_t. \tag{3}$$

For nonnegative costs and unit count value, sorting costs and taking the longest feasible prefix solves this stage in $O(n \log n)$ time. This elementary cardinality construction is used as a reference, not claimed as a new optimization algorithm. A fixed, outcome-blind secondary score $q_i$ then determines the composition:

$$\max_z \sum_i z_i q_i \quad \text{subject to} \quad \sum_i z_i = K_t^*, \quad \sum_i z_i \widehat{c}_{it} \leq B_t, \quad z_i \in \{0,1\}. \tag{4}$$

The equality prevents the secondary objective from sacrificing feasible count *at the current policy state*. It does not force distinct policies to acquire identical cumulative counts after their histories diverge. Both stages use only currently available features and predictions, never unobserved candidate outcomes or test labels.

### 3.3 Acquisition policies

**CountOnly-MinRisk** minimizes predicted risk among maximum-count sets by setting $q_i = -\widehat{c}_i$. It is abbreviated MinRisk in tables. **LRBE-Uncertainty** uses $q_i = \widehat{p}_i(1 - \widehat{p}_i)$. **LRBE-FAVE** uses the frozen FAVE-v2.1 score, a feature-space local information proxy (Appendix A). Its Fisher-style geometry is a surrogate constructed from predictions and covariates, not the Hessian of the fitted boosted-tree model or a guarantee of retraining improvement.

**LRBE-IG** uses empirical committee mutual information. At each round, 20 models are fitted to same-size bootstrap samples of the current visible set. If $p_i^{(r)}$ is member $r$'s default probability and $H(p) = -p \log p - (1-p)\log(1-p)$, then

$$q_i^{\mathrm{IG}} = H\left(\frac{1}{20}\sum_{r=1}^{20} p_i^{(r)}\right) - \frac{1}{20}\sum_{r=1}^{20} H\left(p_i^{(r)}\right). \tag{5}$$

Committee members use the canonical XGBoost hyperparameters and only visible labels. The committee determines acquisition scores only; the single canonical model continues to define candidates, predicted costs, and final evaluation. Scores add across selected applicants, so this comparator is a single-item IG sum rather than joint BatchBALD. **QualityFirst-IG** maximizes the same nonnegative score under the risk constraint without the count equality. IG is an independent acquisition baseline, not a claimed contribution.

**Quality-first policies** maximize the corresponding nonnegative uncertainty or FAVE utility under the risk constraint without the count equality. **Joint-Best-Val-Uncertainty** maximizes $\sum_i z_i(\bar{q}_i - \lambda \bar{c}_i)$ over positive-score candidates, where bars denote within-pool min–max normalization. Its parameter is chosen using only the initial visible set, split into 60% training, 20% pseudo-candidates and 20% validation. Validation PR-AUC determines the choice, with Recall and then smaller $\lambda$ as tie-breakers. These are diagnostic comparators for the acquisition objectives, not implementations of all prior exploration algorithms.

Two additional controls vary randomization. **RiskBounded-Random** visits candidates in a reproducible pseudo-random order and accepts each if its cost fits the remaining budget. **RiskBounded-Epsilon** instead preserves $K_t^*$ and uses $q_i = -0.9\bar{c}_i + 0.1u_i$, where $u_i$ is a deterministic hash-based random score. We retain its archive name but call it the *random-score* control in prose: 0.1 is an objective weight, not a 10% random-approval probability or conventional epsilon-greedy rule. **NoExplore** reveals ordinary approvals only. **Oracle** also reveals every candidate outcome and ignores the exploration budget; it is a full-feedback diagnostic, not a guaranteed upper bound on test performance.

## 4 Experimental Design

### 4.1 Data and replay protocol

For each seed, a random permutation assigns 25% of records to a fixed outer test set. The first 20% of the remaining stream form the initial visible set, with

**Table 1.** Exact input cohorts and exposure semantics. Raw risk units are not comparable across datasets.

| Dataset | Records | Exposure and evidence role |
|---|---|---|
| HMEQ | 5,960 | LOAN amount; development |
| LendingClub | 16,566 | loan-amount proxy; external validation |
| UCI | 30,000 | unit exposure; development |
| PPDai | 55,596 | unit exposure; exploratory external |

a minimum of 100 records; all remaining records are split into ten incoming batches. The initial visible set is therefore a random cohort sample rather than the output of a historical approval rule. Policy-dependent selective observation begins only when the policies process the ten subsequent batches: ordinary approvals are observed automatically, whereas exploratory approvals are chosen from the threshold-defined candidate set. Thus the experiment is *random-order offline replay*, not historical-time evaluation.

Fresh confirmation seeds 1001–1010 are disjoint from development seeds 0–19. The fresh design includes an objective comparison at $\rho = 0.1$, a four-policy sweep over $\rho \in \{0.025, 0.05, 0.1, 0.2, 0.4\}$, and NoExplore/Oracle references. The two randomization controls use the same fresh seeds at $\rho = 0.1$. A separate primary IG block at $\rho = 0.1$ reruns MinRisk, LRBE-FAVE, QualityFirst-FAVE, LRBE-IG, and QualityFirst-IG together, yielding 200 dynamic episodes. Historical control outputs are not spliced into that block. Development analyses are retained separately because their protocols differ from the fresh uncertainty estimates. All reported analyses use the registered seeds, policies, and archived raw outputs.

A controlled one-step experiment uses synthetic tasks with prespecified seeds 2001–2020. Each task contains a selectively observed initial set, one incoming batch, and separate test strata for an initially underrepresented region and its complement. It crosses unit exposure with normalized lognormal exposure and evaluates region-1 weights of 0.1 and 0.5. MinRisk, LRBE-IG, QualityFirst-IG, and a maximum-count random-score selector share the fitted pre-acquisition state, candidates, costs, numerical budget, and branch-training seed. The experiment measures immediate count sacrifice and predicted-risk use, followed by one retraining step; it tests specified condition interactions rather than long-run policy behavior.

### 4.2 Model, optimization and observation controls

All fresh policies use XGBoost with 100 trees, maximum depth 3, learning rate 0.1, binary logistic objective, and histogram-based training. Numeric features receive median imputation and standardization; categorical features receive a missing-value category and one-hot encoding. Target, ID and the designated exposure column are excluded from predictors. Preprocessing is fitted on visible

records only. Each round fits a pre-acquisition model for selection and a post-acquisition model for test evaluation. The outer test set never selects parameters or candidates; population default rates and selected outcomes are used only in offline diagnostics.

Maximum-count selection uses a sorted-cost prefix with a $10^{-10}$ numerical allowance. The common second-stage binary program and quality-first/joint programs use SciPy's mixed-integer solver; an unsuccessful solve raises an error. Risk auditing permits a $10^{-6}$ absolute allowance. Candidate costs are strictly positive under the validated exposures and rejection threshold, so zero budgets yield zero feasible count. Empty candidate sets also yield no exploratory selections. GPU training and supported FAVE matrix operations accelerate computation, while optimization remains CPU-side. The registered bootstrap committee runs on CPU with four concurrent workers; the canonical model remains on GPU. Appendix A records implementation and validation details.

### 4.3 Outcomes and statistical comparisons

We report cumulative exploratory count, cumulative predicted risk, final-test default Recall, and Brier score. Recall is $TP/(TP + FN)$ at $\tau = 0.5$; Brier is the mean squared error of default probabilities. The acquired default fraction describes the selected outcomes. A separate feedback diagnostic, $F$, is the default fraction in the entire visible set divided by the default fraction in the non-test stream. It therefore includes ordinary approvals and the initial set, not only exploratory selections. Neither $F$ nor mean selected predicted probability establishes exact applicant overlap or an independent composition effect.

At each dataset and budget ratio, final policy differences are paired over seeds. Tables show mean differences and pointwise 95% percentile bootstrap intervals from 10,000 seed resamples. Counts that are close descriptively are not called experimentally matched; intervals crossing zero are not equivalence or noninferiority tests. The comparisons are exploratory across multiple metrics and conditions, so isolated interval exclusions are not familywise significance claims.

The IG block adds a same-state mechanism audit. At each round of the MinRisk trajectory, all five selectors receive the same visible set, candidates, cost vector, and numerical budget. We record count sacrifice $D(S) = K_t^* - |S|$, composition premium $P(S) = \widehat{R}(S) - M(|S|)$, where $M(k)$ is the minimum predicted risk of a size-$k$ set, maximum-count slack, selected IDs, and Jaccard overlap with MinRisk. This audit describes immediate selector behavior; because alternative selections do not update the anchor trajectory, it is not a downstream counterfactual.

The controlled experiment applies the same 10,000-draw paired seed bootstrap to LRBE-IG minus QualityFirst-IG count, LRBE-IG minus MinRisk predicted risk and Brier score, and the prespecified exposure-by-region-weight interactions. Its confirmation run holds training randomness fixed across branches. Candidate-label perturbations leave predictions, candidates, IG scores, and selections unchanged; test-label perturbations leave the fitted predictor unchanged.

For feedback–prediction associations, the primary variables are raw gains from the same-seed NoExplore reference: $\Delta F = F_m - F_N$, $\Delta\text{Recall} = \text{Recall}_m - \text{Recall}_N$, and $\Delta\text{Brier} = \text{Brier}_N - \text{Brier}_m$. Positive Brier gain means improvement. Spearman associations are computed separately within dataset and budget, retaining all methods belonging to a resampled seed in each of 10,000 cluster-bootstrap draws. Budget-centered associations are secondary descriptions, not causal estimates. Reference-gap ratios are supplemental only: $(M_m - M_N)/(M_O - M_N)$ is interpretable as recovery only when the oriented Oracle gap is positive; nonpositive or numerically unresolved gaps are N/A. The primary raw-gain analysis avoids dividing by these unstable reference differences.

*Cohort provenance.* UCI is the Default of Credit Card Clients dataset [23]; HMEQ is the home-equity cohort documented in [3]. LendingClub merges the documented DePaul-hosted 2014 and 2015 public samples into a mature Fully Paid/Charged Off cohort [7]. Its outcomes concern historically issued loans, not observed outcomes for genuinely rejected applicants. PPDai is linked to Figshare file 26049578 [9]; the current reconstructed source is recorded by hash, but byte parity with the historical processed file is not established. It therefore carries exploratory evidentiary weight and is not pooled with historical PPDai results. Its exposure column is identically one. Hash-checked source instructions identify the exact inputs without redistributing third-party data. The strict exposure audit finds no missing, nonnumeric, nonfinite or nonpositive exposure in the three stored exposure columns; UCI constructs unit exposure directly.

## 5 Results

### 5.1 The cost of sacrificing current feasible count

The central empirical question is whether a secondary acquisition objective earns the feedback quantity it gives up. Removing the count equality quantifies that cost. Relative to QualityFirst-FAVE, LRBE-FAVE acquires 67.1 [62.6, 71.4] additional labels on HMEQ, 209.6 [202.5, 216.1] on LendingClub, 2.0 [0.9, 3.1] on PPDai, and 56.4 [54.7, 58.2] on UCI. Recall favors LRBE-FAVE on HMEQ by 0.0175 [0.0041, 0.0297]; the other three Recall intervals contain zero. The quantity ordering follows from the current-state count constraint. The empirical result is that giving up this quantity does not yield a consistent predictive improvement across the four datasets.

The random-order control provides a second view. Against MinRisk, it acquires 87.0 fewer labels on HMEQ, 317.6 fewer on LendingClub, and 59.5 fewer on UCI; PPDai differs by 0.3. HMEQ Recall is lower by 0.0268 [-0.0395, -0.0140], while the other Recall intervals contain zero. The fixed-count random-score control instead remains within 2.7 labels of MinRisk in every dataset, and all Recall and Brier intervals contain zero. These controls compare complete policies rather than isolate cardinality from later state changes.

FAVE and MinRisk have close observed cumulative counts, but this is an empirical similarity rather than a matching constraint across policies. FAVE's

**Table 2.** Primary IG mechanism contrasts at $\rho = 0.1$ ($n = 10$ paired seeds). All differences use LRBE-IG as the left policy; entries are means with pointwise 95% intervals.

| Dataset | $\Delta K$ vs. QF-IG | $\Delta \widehat{R}$ vs. MinRisk |
|---|---|---|
| HMEQ | 31.3 [28.3,34.5] | 28,765.5 [19,361,39,425] |
| LendingClub | 49.5 [46.0,53.1] | 27,025.7 [19,238,35,010] |
| UCI | 40.1 [36.8,43.6] | 1.827 [1.004,2.733] |
| PPDai[†] | 0.2 [0.0,0.5] | 1.024 [0.497,1.606] |

[†]Reconstructed-source exploratory evidence with unit exposure; not pooled with historical PPDai results.

count differences are 1.1 [-0.9, 3.0] on HMEQ, −0.1 [-2.5, 2.6] on LendingClub, 0.0 [-1.0, 1.1] on PPDai, and −0.3 [-2.0, 1.5] on UCI. It uses more predicted risk in all four datasets, while every Recall interval contains zero. Three Brier intervals contain zero; UCI favors FAVE by −0.00018 [−0.00028, −0.00008]. This retains the dataset-specific benefit without implying a consistent cross-dataset return.

### 5.2 Independent information gain separates quantity from score choice

The IG extension tests whether the quantity result depends on FAVE's linearized score. All five policies are rerun in a single evidence block at $\rho = 0.1$; Table 2 reports the two preregistered contrasts. Relative to QualityFirst-IG, LRBE-IG acquires 31.3 [28.3, 34.5] additional labels on HMEQ, 49.5 [46.0, 53.1] on LendingClub, and 40.1 [36.8, 43.6] on UCI. PPDai differs by 0.2 [0.0, 0.5] in its small-feedback regime. No corresponding Recall interval excludes zero. Brier intervals also contain zero except on UCI, where LRBE-IG is higher by 0.000156 [0.000019, 0.000298]. Thus preserving $K_t^*$ changes acquired quantity on the three informative-count datasets, but the present experiment does not establish a predictive gain from that additional feedback.

Against MinRisk, LRBE-IG has a close cumulative count but spends more predicted risk on every dataset (Table 2). None of the Recall or Brier intervals excludes zero. The independent score therefore changes composition and risk use without a detectable downstream return in this block.

The same-state audit fixes the MinRisk visible set, candidates, costs, and numerical budget. Across 100 states per dataset, LRBE-IG has zero count sacrifice. QualityFirst-IG gives up 3.46, 5.37, 4.04, and 0.02 applicants per state on HMEQ, LendingClub, UCI, and PPDai, respectively, while changing selection overlap. Because alternative sets do not generate later audit states, this is a mechanism check rather than a downstream counterfactual.

### 5.3 Controlled conditions reproduce the selection trade-off

The one-step experiment holds the acquisition state and training randomness fixed while varying exposure heterogeneity and the evaluation weight of an un-

derrepresented region. LRBE-IG selects 2.5 [1.45, 3.65] more applicants than QualityFirst-IG under unit exposure and 3.2 [1.75, 4.85] more under heterogeneous exposure. Relative to MinRisk, it uses 0.221 [0.165, 0.284] and 0.136 [0.101, 0.171] more predicted-risk units, respectively. None of the four primary Brier intervals excludes zero.

The prespecified condition interactions are also uncertain. Increasing the target-region weight from 0.1 to 0.5 changes the IG-minus-MinRisk Brier contrast by 0.000126 [−0.000005, 0.000255] under unit exposure and −0.000058 [−0.000217, 0.000102] under heterogeneous exposure; the difference between those changes is −0.000184 [−0.000413, 0.000029]. Thus the controlled setting reproduces the immediate quantity–risk trade-off but does not show that these two conditions determine its predictive return. The incoming batch averages about 3,751 ordinary approvals, compared with exploratory $K^*$ means of 29.55 and 118.7, so the evaluated retraining intervention is small relative to the shared ordinary feedback.

### 5.4 Budget sensitivity

The budget ratio has a much larger and more consistent relation with acquired quantity than the secondary score. HMEQ MinRisk grows from 57.4 labels at $\rho = 0.025$ to 374.0 at $\rho = 0.4$; FAVE grows from 58.0 to 375.8. LendingClub MinRisk grows from 207.6 to 1,253.1 and FAVE from 206.3 to 1,253.3. Recall moves less and method ordering changes across budgets. Figure 1 shows representative HMEQ and LendingClub trajectories. The sweep changes both the scale of feedback and the subsequent policy state, so it is a sensitivity analysis rather than a quantity-only intervention.

### 5.5 Feedback diagnostics do not have a single translation pattern

Normalized recovery divides policy changes by an Oracle–NoExplore gap, and is not interpretable as recovery when Oracle is not better. In the present data, the Recall gap is nonpositive in 10 of 40 dataset–seed references: two LendingClub seeds, two PPDai seeds, including one zero, and six UCI seeds. Oracle gaps are also nonpositive in 21 of 40 ROC-AUC and 15 of 40 PR-AUC references. We therefore use raw gains from NoExplore as the primary analysis and reserve normalized recovery for positive-gap cells.

Figure 2 relates visible-set default-ratio gain to Recall and Brier improvement at $\rho = 0.1$. HMEQ has a positive Recall association, $\rho_s = 0.397$ [0.013, 0.660], but its Brier association is uncertain, 0.062 [−0.175, 0.381]. LendingClub, PPDai, and UCI Recall intervals contain zero at this budget, as do all four Brier intervals. Across the five within-budget analyses, HMEQ Recall estimates range from 0.389 to 0.514 and all intervals exclude zero. LendingClub instead has negative Recall associations at $\rho = 0.025$ and 0.05, followed by uncertain estimates. PPDai's smallest-budget cells have near-degenerate ranks and yield perfect coefficients, so those boundary values are not interpreted as general evidence.

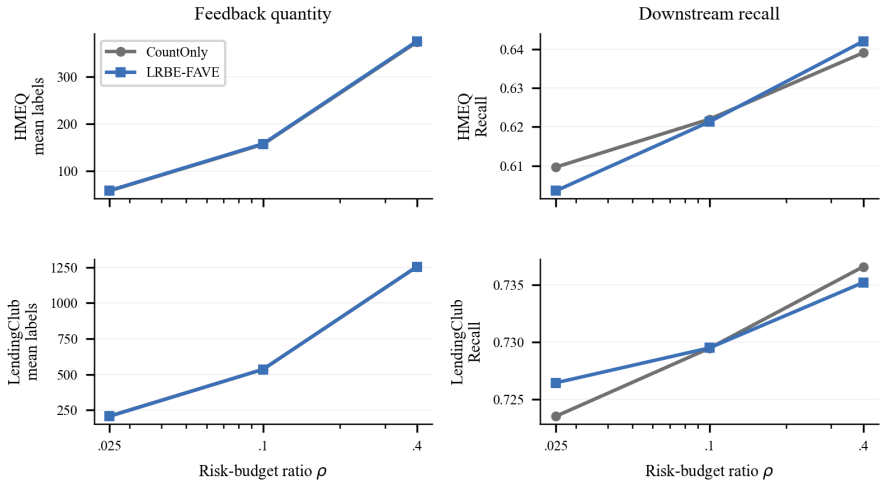



**Fig. 1.** Mean cumulative exploratory count and final Recall across budget ratios for HMEQ and LendingClub (ten fresh seeds). Larger budgets increase feedback, while MinRisk and FAVE do not have a uniform Recall ordering.

These associations concern a diagnostic of the complete visible set, which includes the initial set and ordinary approvals. A larger ratio is not necessarily closer to the population target, and policy-specific predictors make selected mean PD an unsuitable cross-policy identity measure. Feedback diagnostics align with Recall in some regimes but are not a stable proxy across datasets, budgets and outcomes. The correlations do not estimate a causal composition effect.

### 5.6 Predicted feasibility and realized default exposure

All non-Oracle selections obey the fitted point-cost constraint within audit tolerance. Across 104 equally sized method–dataset–budget groups, the mean rate at which realized defaulted exposure exceeds the policy's numerical predicted budget is 0.4798, with group rates ranging from 0.01 to 0.84. Because each group contains 100 seed–round observations, the unweighted group mean also equals the observation-weighted mean for these included cells. The range is descriptive, not a confidence interval.

This comparison evaluates an ex-post loss exceedance against a budget constructed from fitted expected costs, rather than against a realized-loss quantile. The exceedance rate therefore shows that point-estimate feasibility does not ensure ex-post loss-cap compliance. It does not separately identify model miscalibration, stochastic default variation, exposure assumptions, or dependence among loans.

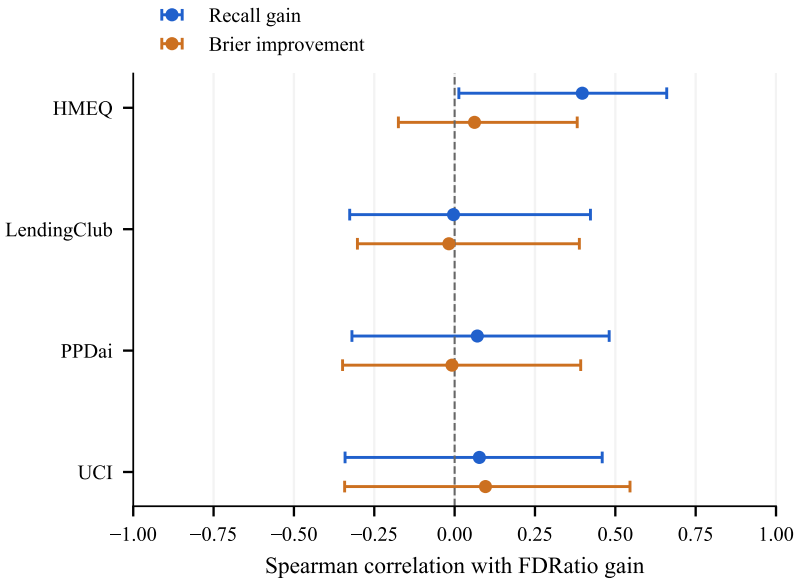



**Fig. 2.** Within-dataset associations between visible-set default-ratio gain and downstream improvement at $\rho = 0.1$. Points use 40 terminal policy cells (four policies, ten seeds); bars are pointwise 95% intervals from 10,000 whole-seed bootstrap resamples. Positive Brier gain means lower error.

*Descriptive computation time.* Recorded episode wall time includes training, selection, evaluation, and system effects. Mean LendingClub times in the IG block are 44.0 s for MinRisk, 82.8 s for LRBE-FAVE, and 498.6 s for LRBE-IG; these are end-to-end episode descriptions rather than isolated acquisition latency.

## 6 Discussion and Scope

*What a quantity-preserving reference contributes.* Maximizing feasible count makes the opportunity cost of a secondary acquisition score observable. The scientific question begins after that arithmetic fact: does the score earn the observations it displaces? FAVE, IG, and the random-order control all produce sizeable quantity differences when the maximum-count equality is removed, yet composition-first selection has no consistent predictive advantage. Conversely, the additional labels preserved by LRBE do not reliably improve prediction; UCI Brier is slightly worse for LRBE-IG than for QualityFirst-IG. The useful outcome is therefore a reference comparison, not a universal ordering. Within the eligibility rule and predicted-risk constraint, MinRisk states the least predicted risk required to retain the current maximum count.

*Policy effects and composition evidence.* The three experimental views answer different questions. Dynamic replay compares complete acquisition policies that begin with the same data stream but evolve through policy-specific feedback, predictions, candidates, and budgets. Same-state audits fix those quantities and show exactly how selectors exchange count, risk, and score at one decision point. The controlled experiment adds a single retraining step after fixing the acquisition state. It reproduces the count and predicted-risk contrasts, while its exposure and target-weight interactions remain uncertain. Taken together, the results locate the trade-off at selection but do not reduce long-run policy differences to composition alone.

*Why predictive differences can remain small.* Most labels added in the controlled task come from ordinary approvals shared across selectors: about 3,751 per incoming batch, compared with exploratory $K^*$ means of 29.55 under unit exposure and 118.7 under heterogeneous exposure. Differences among exploratory selections therefore modify only a fraction of the post-acquisition training set. This scale contrast offers one plausible account of the small Brier differences, but the experiment was designed to test exposure and target-weight interactions rather than isolate dilution. More broadly, an acquisition score is useful only when the labels it changes affect the fitted predictor and the target metric; improved score totals or altered feedback composition are intermediate outcomes.

*Scope of inference.* The evidence concerns random-order replay on historical cohorts with observed outcomes. LendingClub does not contain outcomes for actual historical rejects, labels arrive immediately in the simulation, and $\tau = 0.5$ is a common experimental operating point rather than a production cutoff. PPDai contributes unit-exposure exploratory evidence; monetary heterogeneity comes from HMEQ and LendingClub. The IG comparison uses the registered primary budget $\rho = 0.1$, while the broader budget sweep uses the original policy set. Published online exploration methods motivate the question, but the randomized controls are not implementations of those algorithms. The conclusions apply to the evaluated acquisition policies and replay conditions.

*Risk interpretation.* An optimizer can enforce a sum of predicted losses without controlling the tail of realized loss. Here, an exceedance means that realized defaulted exposure in a seed–round observation is larger than the policy's point-estimate budget. It is an ex-post comparison with an expected-loss quantity, not a direct estimate of safety failure. The observed frequency does not attribute the difference separately to probability error, default randomness, or exposure assumptions. Live credit experimentation would require product-specific exposure and recovery definitions, delayed-outcome handling, concentration and fairness safeguards, and explicit stopping rules. None of these operational guarantees follows from the current point-cost constraint.

## 7 Conclusion

Risk-budgeted feedback acquisition involves two distinct choices: how many outcomes to make observable and which applicants should provide them. LRBE separates these choices by preserving the current maximum feasible count before applying a composition score. Across FAVE, independent IG, and randomized controls, removing that constraint often sacrifices substantial feedback, but preserving more labels does not itself guarantee better prediction. At nearly the same cumulative count, FAVE and IG also spend more predicted risk than MinRisk without a consistent downstream return. The controlled analysis reproduces these selection trade-offs but does not identify exposure heterogeneity or target-region emphasis as reliable moderators. Acquisition studies should therefore report quantity, predicted-risk use, composition change, and downstream performance together. Minimum-risk maximum-count selection provides the reference needed to decide whether a more elaborate score earns its cost.

## A Secondary Score and Reproducibility

*Frozen FAVE-v2.1 score.* Let $v_j = [1, x_j^\top]^\top$ be the augmented, preprocessed feature vector. For the current visible set $\mathcal{L}_t$ and label-free reference batch $\mathcal{U}_t$, define

$$H_t = \sum_{j \in \mathcal{L}_t} \max\{\widehat{p}_j(1-\widehat{p}_j), 10^{-6}\} v_j v_j^\top, \tag{6}$$

$$G_t = |\mathcal{U}_t|^{-1} \sum_{j \in \mathcal{U}_t} \widehat{p}_j v_j v_j^\top, \qquad \eta = \max\{10^{-6}\,\mathrm{tr}(H_t)/d, 10^{-10}\}, \tag{7}$$

$$A_t = H_t + \eta I, \qquad \widetilde{w}_i = \max\{\widehat{p}_i(1-\widehat{p}_i)\widehat{p}_i^{0.5}, 10^{-10}\}, \tag{8}$$

$$q_i^{\mathrm{FAVE}} = \frac{\widetilde{w}_i v_i^\top A_t^{-1} G_t A_t^{-1} v_i}{1 + \widetilde{w}_i v_i^\top A_t^{-1} v_i}. \tag{9}$$

Here $d$ includes the intercept coordinate. Predicted probabilities are clipped to $[10^{-9}, 1-10^{-9}]$ before these operations; nonfinite score outputs are zeroed and scores are clipped below at zero. The default-weighting exponents and numerical regularization are fixed. This linearized feature-space construction is used with boosted-tree predictions as a scoring proxy; it is not an exact influence calculation for the tree-fitting procedure. The Joint comparator selects $\lambda \in \{0.01, 0.1, 0.5, 1, 2, 5, 10\}$ using only initial-set validation. The initial split is stratified where possible, with a deterministic random split as fallback.

*Input and trajectory audit.* Exact-input SHA-256 checks cover HMEQ, LendingClub and PPDai. All 78,122 records have numeric, finite, positive exposure; PPDai values are all one. UCI constructs ones directly. The strict input loader rejects invalid exposure rather than replacing it by zero. Eight validation trajectories (four datasets, seed 1001, MinRisk and FAVE) agree with the archived outputs in the checked counts, risk values, and predictive metrics. This check is not

an additional statistical replication. Of 40 paired post-initial rounds, incoming-batch hashes agree in all 40, while candidates and numerical budgets agree in only four. This directly documents the policy-specific state evolution. The IG and controlled-condition archives retain their own registered seeds and are not pooled as independent replications. The controlled confirmation uses common branch randomness and label-perturbation checks; all 320 endpoint rows match the original block in selection, risk, Recall, and Brier score. Source hashes, full-precision tables, parameter grids, and executable tests accompany the archive.

## References


1. Agrawal, S., Devanur, N.R.: Linear contextual bandits with knapsacks. In: Advances in Neural Information Processing Systems. vol. 29 (2016)
2. Amani, S., Alizadeh, M., Thrampoulidis, C.: Linear stochastic bandits under safety constraints. In: Advances in Neural Information Processing Systems. vol. 32 (2019)
3. Baesens, B., R"osch, D., Scheule, H.: Credit Risk Analytics: Measurement Techniques, Applications, and Examples in SAS. Wiley (2016). https://doi.org/10.1002/9781119449560
4. Banasik, J., Crook, J.: Reject inference, augmentation, and sample selection. European Journal of Operational Research **183**(3), 1582–1594 (2007). https://doi.org/10.1016/j.ejor.2006.06.072
5. Das, N., Sharma, M., Nanavati, P., Shiragur, K., Deshpande, A.: Cost efficient fairness audit under partial feedback (2025), https://arxiv.org/abs/2510.03734, regML 2025 poster
6. De-Arteaga, M., Dubrawski, A., Chouldechova, A.: Learning under selective labels in the presence of expert consistency. arXiv preprint arXiv:1807.00905 (2018), https://arxiv.org/abs/1807.00905
7. DePaul University Economics Data Archive: Lendingclub bank loan data, 2007–2015 (2015), https://bigblue.depaul.edu/jlee141/econdata/LendingClub_LoanData/, the registered study input is a leakage-audited mature cohort reconstructed from the 2014 and 2015 public samples; its exact processed SHA-256 is recorded in the project registry
8. Donmez, P., Carbonell, J.G.: Proactive learning: Cost-sensitive active learning with multiple imperfect oracles. In: Proceedings of the 17th ACM Conference on Information and Knowledge Management. pp. 619–628 (2008). https://doi.org/10.1145/1458082.1458165
9. Figshare: Frozen ppdai source object, file 26049578 (2020), https://ndownloader.figshare.com/files/26049578, file-level source recorded by the frozen project; incomplete item metadata is one reason PPDai is lower-confidence exploratory evidence
10. Jiang, H., Jiang, Q., Pacchiano, A.: Learning the truth from only one side of the story. In: Proceedings of the 24th International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 130, pp. 2413–2421 (2021), https://proceedings.mlr.press/v130/jiang21b.html
11. Keswani, V., Mehrotra, A., Celis, L.E.: Fair classification with partial feedback: An exploration-based data collection approach. In: Proceedings of the 41st International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 235, pp. 23547–23576 (2024), https://proceedings.mlr.press/v235/keswani24a.html

12. Kilbertus, N., Gomez Rodriguez, M., Sch"olkopf, B., Muandet, K., Valera, I.: Fair decisions despite imperfect predictions. In: Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 108, pp. 277–287 (2020), https://proceedings.mlr.press/v108/kilbertus20a.html
13. Kleinberg, J., Lakkaraju, H., Leskovec, J., Ludwig, J., Mullainathan, S.: Human decisions and machine predictions. The Quarterly Journal of Economics **133**(1), 237–293 (2018). https://doi.org/10.1093/qje/qjx032
14. Lakkaraju, H., Kleinberg, J., Leskovec, J., Ludwig, J., Mullainathan, S.: The selective labels problem: Evaluating algorithmic predictions in the presence of unobservables. In: Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 275–284 (2017). https://doi.org/10.1145/3097983.3098066
15. Marshall, A., Tang, L., Milne, A.: Variable reduction, sample selection bias and bank retail credit scoring. Journal of Empirical Finance **17**(3), 501–512 (2010). https://doi.org/10.1016/j.jempfin.2009.12.003
16. Pacchiano, A., Ghavamzadeh, M., Bartlett, P., Jiang, H.: Stochastic bandits with linear constraints. In: Proceedings of the 24th International Conference on Artificial Intelligence and Statistics. Proceedings of Machine Learning Research, vol. 130, pp. 2827–2835 (2021), https://proceedings.mlr.press/v130/pacchiano21a.html
17. Pacchiano, A., Singh, S., Chou, E., Berg, A.C., Foerster, J.: Neural pseudo-label optimism for the bank loan problem. In: Advances in Neural Information Processing Systems. vol. 34 (2021), https://papers.nips.cc/paper_files/paper/2021/hash/33d6548e48d4318ceb0e3916a79afc84-Abstract.html
18. Scarone, B., Baeza-Yates, R.: The illusion of improvement: Reject inference strategies in credit scoring. arXiv preprint arXiv:2606.18479 (2026), https://arxiv.org/abs/2606.18479
19. Tomanek, K., Hahn, U.: A comparison of models for cost-sensitive active learning. In: Proceedings of COLING 2010: Posters. pp. 1247–1255 (2010), https://aclanthology.org/C10-2143/
20. Wei, D.: Decision-making under selective labels: Optimal finite-domain policies and beyond. In: Proceedings of the 38th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 139, pp. 11035–11046 (2021), https://proceedings.mlr.press/v139/wei21a.html
21. Wu, Y., Shariff, R., Lattimore, T., Szepesvári, C.: Conservative bandits. In: Proceedings of the 33rd International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 48, pp. 1254–1262 (2016), https://proceedings.mlr.press/v48/wu16.html
22. Yang, Y., Liu, Y., Naghizadeh, P.: Adaptive data debiasing through bounded exploration. In: Advances in Neural Information Processing Systems. vol. 35 (2022). https://doi.org/10.52202/068431-0111, https://proceedings.neurips.cc/paper_files/paper/2022/hash/0a166a3d98720697d9028bbe592fa177-Abstract-Conference.html
23. Yeh, I.C.: Default of credit card clients (2016). https://doi.org/10.24432/C55S3H